Subjective Expected Utility

Primitives:

$\mathbb{X}$ a nonempty set of outcomes (not necessarily finite).
$\mathcal{S} = \{1, ..., S\}$, a set of *states*.
$\mathbb{X}^S$, the set of *acts*.[1]
$\succsim \subset \mathbb{X}^S \times \mathbb{X}^S$, a binary relation on $\mathbb{X}^S$ (with $\succ$ and $\sim$ defined in the usual way).

There is no mention of probabilities. The uncertainty is modeled as a state space, $\mathcal{S}$. An *act* $x = (x_1, ..., x_S) \in \mathbb{X}^S$ specifies a consequence x_s in for each state $s \in \mathcal{S}$, and preferences are defined on the set of such acts. A major question is this: *can preferences over acts reveal something about the beliefs of the person over the states*? If $\mathbb{X}^S$ is a connected subset of a Euclidean space and $\succsim \subset \mathbb{X}^S \times \mathbb{X}^S$ is complete, transitive, and continuous, then there is a continuous representation U of $\succsim$ (Debreu, 1959, *Theory of Value*, Section 4.6). But most of the profession uses a *subjective expected utility* (SEU) representation.

Definition (Subjective Expected Utility)**.** *A function $U : \mathbb{X}^S \to \mathbb{R}$ is a* ***subjective expected utility (SEU) representation*** *of $\succsim \subset \mathbb{X}^S \times \mathbb{X}^S$ if there is a real-valued function u on $\mathbb{X}$ and a probability distribution $\pi = (\pi_1, ..., \pi_S)$ on $\mathcal{S}$ such that*

$$U(x) = \sum_{s=1}^{S} \pi_s u(x_s)$$

where $x = (x_1, ..., x_S) \in \mathbb{X}^S$.

The person's preferences over acts are *as if* the person assigns probability π_s to state s. Specializing to $\mathbb{X} = \mathbb{R}_+$, concavity of u in the representation is equivalent to risk aversion: $x \precsim \mathbb{E}[x]\mathbf{e}$, for every $x \in \mathbb{X}^S$, where $\mathbf{e}$ is a vector of 1s.[2] If u is differentiable, then the marginal rate of substitution between money in states s and σ is

$$MRS_{s\sigma}(x) = \frac{\partial U/\partial x_s}{\partial U/\partial x_\sigma} = \frac{\pi_s u'(x_s)}{\pi_\sigma u'(x_\sigma)}$$

for any pair of states s and σ. If $x_s = x_\sigma$, then $MRS_{s\sigma}(x) = {}^{\pi_s}/_{\pi_\sigma}$, giving a sense in which preferences $\succsim$ over acts *reveal* a person's beliefs about the states.

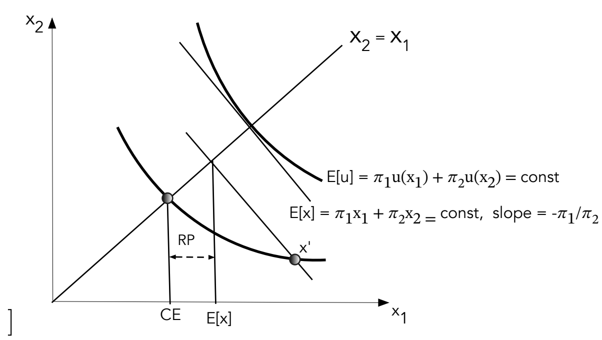

Figure 1: Preferences over acts for $S = 2$: risk aversion implies preferences over monetary acts are convex. For the act x', the certainty equivalent is CE and the risk premium is $RP = [\pi_1 x'_1 + \pi_2 x'_2] - CE$.

[1] In Savage (1954), an act is a function from the set of states, $\mathcal{S}$ into consequences, X. Since I have assumed a finite number of states I can describe and act as a vector $x = (x_1, ..., x_S)$, where x_s is the consequence that the act assigns to state s.

[2] In a slight abuse of notation, $\mathbb{E}[x] = \sum_{s=1}^{S} \pi_s x_s$.

The next figure depicts indifference curves for four people ordered by risk aversion: an increase in risk aversion implies that preferences are "more convex." One interpretation is that, as risk aversion increases, the set of risky acts that a person would accept starting from a riskless act shrinks. If u_1 is more concave than u_0, then

$$MRS_{12}^1(x) = \frac{T'(u_0(x_1))}{T'(u_0(x_2))} \times MRS_{12}^0(x)$$

for some concave function T with $T' > 0$, so $MRS_{12}^1(x) \gtreqless M_{12}^0(x)$ iff $x_2 \gtreqless x_1$. In the limit, as risk aversion becomes infinite, preferences are "max-min," evaluating each act by its worst outcome, without regard for probability: $U(x) = \min\{x_1, x_2\}$. These limit preferences lie outside SEU preferences.

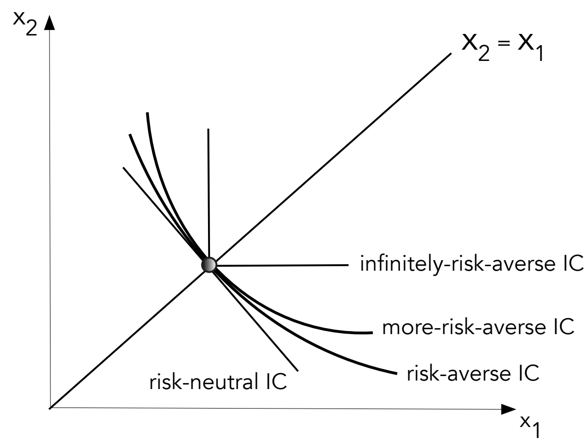

Figure 2: Increases in risk aversion in SEU—and other theories.

As mentioned, SEU preferences are not consistent with the "modal" choice in the Keynes (1921)-Ellsberg (1961) paradox.

In the opening passage of Chapter 7 of his *Theory of Value*, Debreu writes

> The analysis is extended... to the case where uncertain events determine the consumption sets, the production sets, and the resources of the economy. A contract for the transfer of a commodity now specifies, in addition to its physical properties, its location and its date, an *event* on the occurence of which the transfer is conditional. This *new* definition of a commodity allows one to obtain a theory of uncertainty *free from any probability concept* and formally identical with the theory of certainty developed in the preceding chapters [Debreu, 1959, p. 98, emphases added].

Debreu, writing some four years after Savage's *Foundations*, and two years before Ellsberg's paper, emphasizes that his choice theory is *probability-free.*

Debreu strongly suggests we should use this general represenation—which does not imply expected utility or even *probabilistic sophistication* (that a person's choices are consistent with some belief about the states).[3] As mentioned, almost the entire profession uses a subjective expected utility representation in applications.

[3]As do Hirshleifer, Jack. "Investment Decision under Uncertainty: choice—theoretic approaches." *The Quarterly Journal of Economics* 79.4 (1965): 509-536; Yaari, Menahem E. "Some remarks on measures of risk aversion and on their uses." *Journal of Economic Theory* 1.3 (1969): 315-329.; and Machina, Mark J. "'Expected utility/subjective probability'analysis without the sure-thing principle or probabilistic sophistication." *Economic Theory* 26.1 (2005): 1-62.